

Supplementary Online Content

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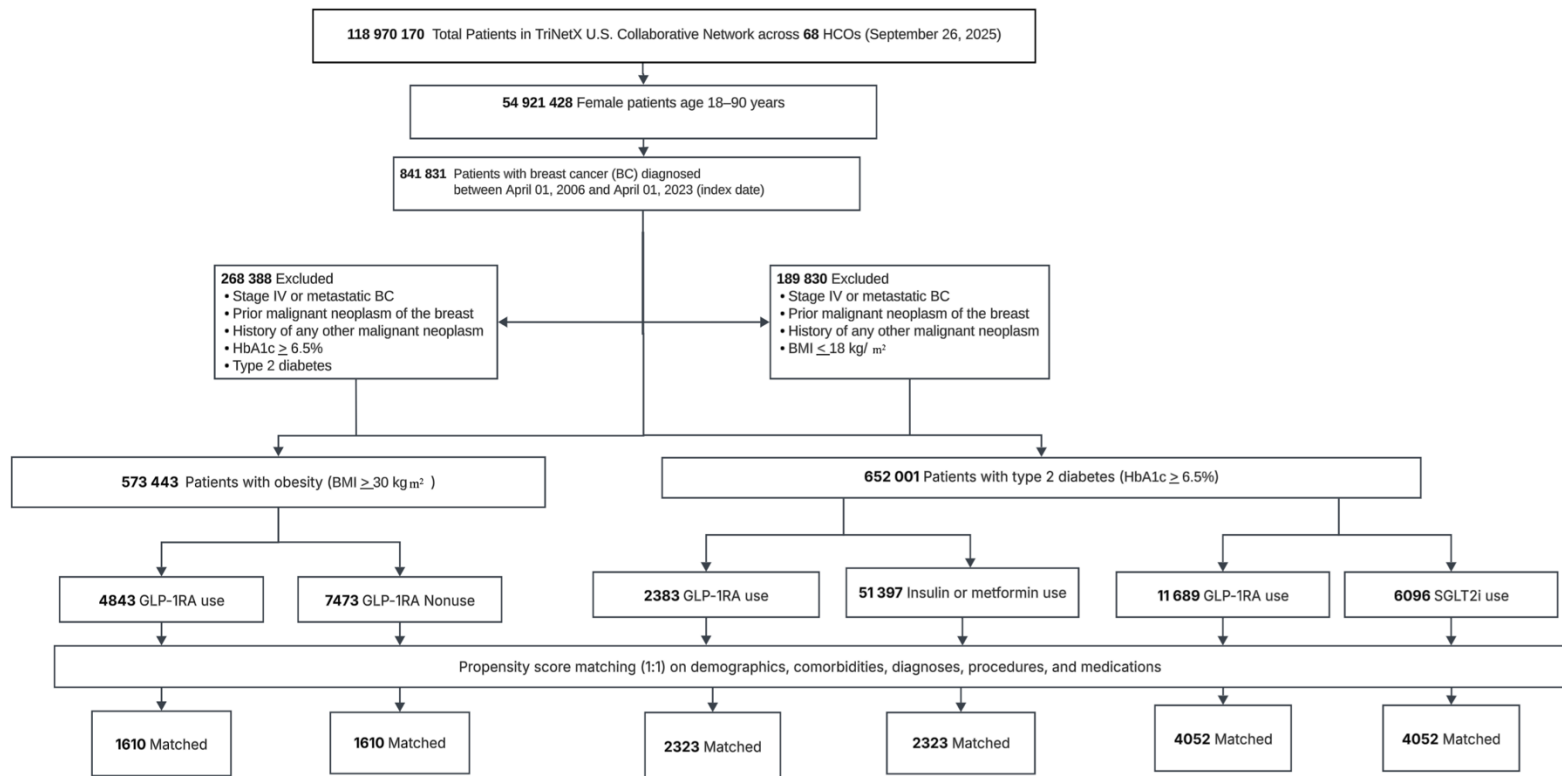
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This supplementary material has been provided by the authors to give readers additional information about their work.

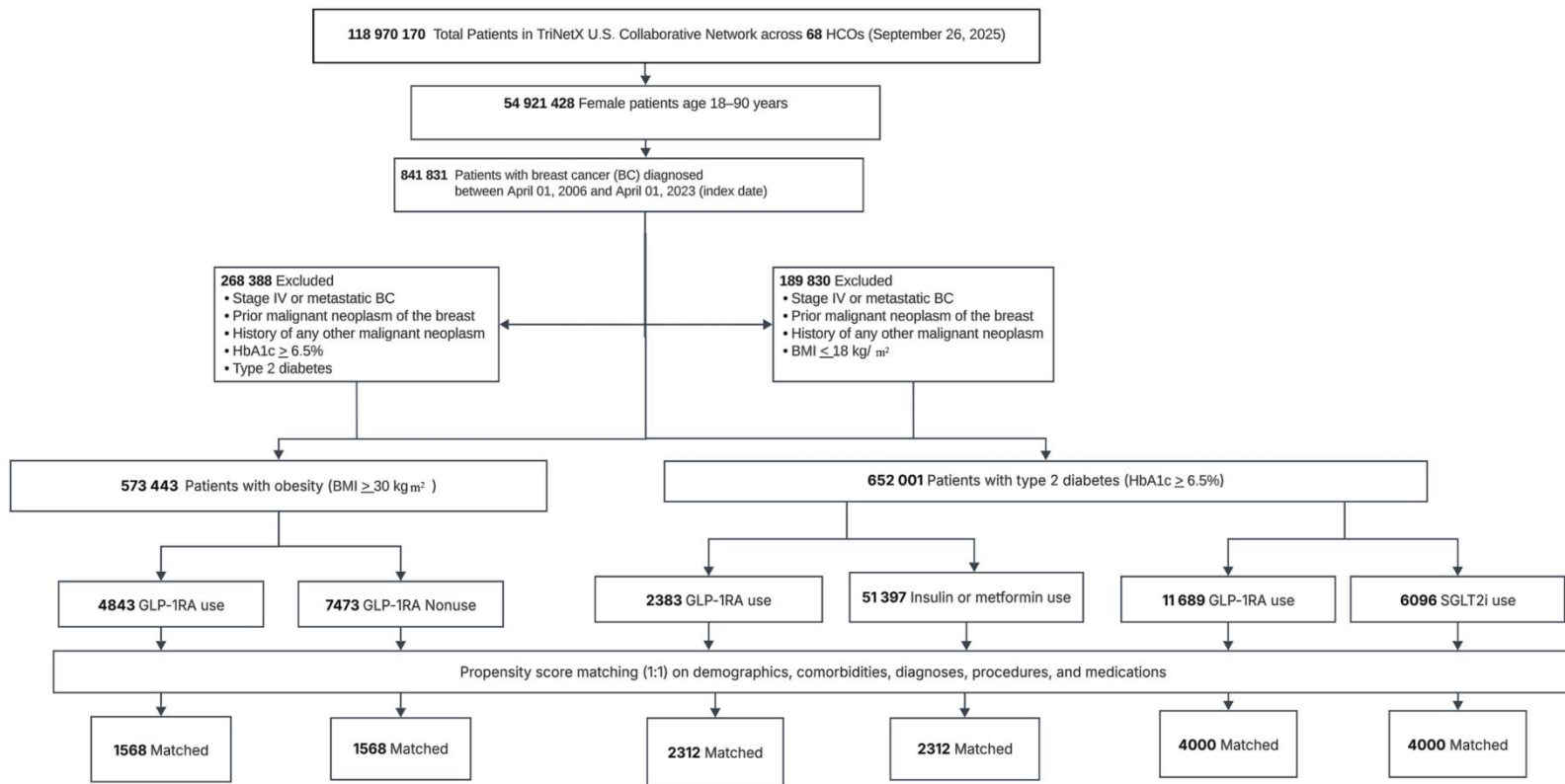
eFigure 1. Study Group Flow Diagram for All-Cause Mortality



Abbreviations: GLP-1RA, glucagon-like peptide-1 receptor agonist; SGLT2i, sodium-glucose cotransporter 2 inhibitors.

The combined total of patients is not necessarily the sum of the individuals across groups because some patients may have been prescribed both GLP-1RAs and insulin, metformin, or SGLT2i during the study period.

eFigure 2. Study Group Flow Diagram for Recurrence-Free Survival



Abbreviations: GLP-1RA, glucagon-like peptide-1 receptor agonist; SGLT2i, sodium-glucose cotransporter 2 inhibitors.

The combined total of patients is not necessarily the sum of the individuals across groups because some patients may have been prescribed both GLP-1RAs and insulin, metformin, or SGLT2i during the study period.

eTable 1. Code Descriptions of Patient Characteristics, Inclusion, and Exclusion Criteria

Variables	Code definition
Socioeconomic risks	ICD-10-CM: Z55–Z65
Comorbidities	
Alcohol related disorders	ICD-10-CM: F10
Nicotine dependence	ICD-10-CM: F17
CKD	ICD-10-CM: N18
COPD	ICD-10-CM: J44
Cerebrovascular diseases	ICD-10-CM: I60–I69
Ischemic heart diseases	ICD-10-CM: I20–I25
Heart failure	ICD-10-CM: I50
Diseases of arteries, arterioles, capillaries	ICD-10-CM: I70–I79
Diseases of liver	ICD-10-CM: K70–K77
Hypertensive diseases	ICD-10-CM: I10–I1A
Disorders of lipoprotein metabolism	ICD-10-CM: E78
Obstructive sleep apnea	ICD-10-CM: G47.33
Overweight and Obesity	ICD-10-CM: E66
Bariatric Surgery Status	ICD-10-CM: Z98.84
Diagnosis	
Carcinoma in situ of breast	ICD-10-CM: D05
ER+ status	ICD-10-CM: Z17.0
ER- status	ICD-10-CM: Z17.1
Stage 0	AJCC: S0
Stage 1	AJCC: S1
Stage 2	AJCC: S2
Stage 3	AJCC: S3
HER2-	TriNetX curated code: 3850011
HER2+	TriNetX curated code: 3850012
Medications	
Metformin	RxNorm: 6809
SGLT2i	ATC: A10BK
Procedures	
Mastectomy Procedures ^a	CPT: 1015054
Mastectomy, partial	CPT: 19301
Chemotherapy	TriNetX curated code: 1002
Treatment	
Tamoxifen	RxNorm: 10324
Cyclophosphamide	RxNorm: 3002
Aromatase inhibitors	ATC: L02BG
Platinum compounds	ATC: L01XA
Anthracyclines	ATC: L01DB
Taxanes	ATC: L01CD
Inclusion and exclusion	
Breast cancer	ICD-10-CM: C50
Stage 4	TriNetX curated code: STAGE S4
M1	TriNetX curated code: TNM_M1
Hemoglobin	TriNetX curated code: 9037
BMI	TriNetX curated code: 9083
Type 2 diabetes mellitus	ICD-10-CM: E11
Malignant neoplasms of bone and articular cartilage	ICD-10-CM: C40–C41

Malignant neoplasms of respiratory and intrathoracic organs	ICD-10-CM: C30-C39
Malignant neoplasms of mesothelial and soft tissue	ICD-10-CM: C45-C49
Malignant neoplasms of urinary tract	ICD-10-CM: C64-C68
Malignant neoplasms of eye, brain and other parts of central nervous system	ICD-10-CM: C69-C72
Malignant neuroendocrine tumors	ICD-10-CM: C7A
Malignant neoplasms of female genital organs	ICD-10-CM: C51-C58
Malignant neoplasms of digestive organs	ICD-10-CM: C15-C26
Secondary malignant neoplasm of respiratory and digestive organs	ICD-10-CM: C78
Malignant neoplasm of other and ill-defined sites	ICD-10-CM: C76
Malignant neoplasms of lymphoid, hematopoietic and related tissue	ICD-10-CM: C81-C96
Malignant neoplasms of thyroid and other endocrine glands	ICD-10-CM: C73-C75
Melanoma and other malignant neoplasms of skin	ICD-10-CM: C43-C44
GLP-1RA	ATC: A10BJ RxNorm: 2601723 (Tirzepatide)
Orlistat	RxNorm: 37925
Phentermine	RxNorm: 8152
Topiramate	RxNorm: 38404
Bupropion	RxNorm: 42347
Setmelanotide	RxNorm: 2469247
Naltrexone	RxNorm: 7243
Metformin	RxNorm: 6809
Insulins and analogues	ATC: A10A
Sodium-glucose co-transporter 2 inhibitors	ATC: A10BK
Outcome Definitions	
Deceased for any cause	TriNetX curated code: R99
Recurrence	
Secondary malignancy of lymph nodes	ICD-10-CM: C77.x
Secondary malignancy of respiratory/digestive organs	ICD-10-CM: C78.x
Secondary malignancy of other specified sites, including bone, brain, skin, and breast	ICD-10-CM: C79.0–C79.9, C79.81

Abbreviations: GLP-1RA, glucagon-like peptide-1 receptor agonist; SGLT2i, sodium-glucose cotransporter 2 inhibitors, ER+, estrogen receptor positive; ER-, estrogen receptor negative; HER2+, Human epidermal growth factor receptor 2-positive; HER2-, Human epidermal growth factor receptor 2-negative; CKD, chronic kidney disease; COPD, other chronic obstructive pulmonary disease; OSA, obstructive sleep apnea.

*Mastectomy procedures code 1015054 is a TriNetX code referencing a unique identifier within the NIH Value Set Authority Center system for mastectomy procedures.

eTable 2. Baseline Characteristics of GLP-1RA Use vs Nonuse Before and After Propensity Score Matching for All-Cause Mortality

Characteristic	Before PSM ^a			After PSM ^a		
	No. (%)		SMD ^b	No. (%)		SMD ^b
	GLP-1RA (n=4843)	Nonuse (n=7473)		GLP-1RA (n=1610)	Nonuse (n=1610)	
Age at Index, mean (SD), y	60.1 (10.7)	54.5 (11.4)	0.508	57.7 (10.7)	58.6 (11.3)	0.078
Race						
American Indian or Alaska Native	10 (0.2)	24 (0.3)	0.022	10 (0.6)	10 (0.6)	0.001
Asian	94 (1.9)	62 (0.8)	0.095	24 (1.5)	20 (1.2)	0.021
Black or African American	731 (15.1)	1044 (14.0)	0.032	235 (14.6)	242 (15.0)	0.012
Native Hawaiian or Other Pacific Islander	15 (0.3)	23 (0.3)	0.001	10 (0.6)	10 (0.6)	0.001
Other ^c	143 (3.0)	225 (3.0)	0.003	60 (3.7)	59 (3.7)	0.003
Unknown race	269 (5.6)	411 (5.5)	0.002	84 (5.2)	94 (5.8)	0.027
White	3582 (74.0)	5684 (76.1)	0.048	1199 (74.5)	1188 (73.8)	0.016
Ethnicity						
Hispanic or Latino	305 (6.3)	468 (6.3)	0.001	101 (6.3)	100 (6.2)	0.003
Not Hispanic or Latino	3767 (77.8)	5576 (74.6)	0.074	1181 (73.4)	1203 (74.7)	0.031
Unknown Ethnicity	771 (15.9)	1429 (19.1)	0.084	328 (20.4)	307 (19.1)	0.033
Socioeconomic risks ^d	78 (1.6)	72 (1.0)	0.057	23 (1.4)	19 (1.2)	0.022
Lifestyle factors						
Nicotine dependence	133 (2.7)	336 (4.5)	0.094	55 (3.4)	46 (2.9)	0.032
Alcohol related disorders	37 (0.8)	129 (1.7)	0.087	19 (1.2)	20 (1.2)	0.006
Breast cancer characteristics						
Carcinoma in situ of breast	403 (8.3)	751 (10.0)	0.06	119 (7.4)	109 (6.8)	0.024
ER+ status	935 (19.3)	825 (11.0)	0.232	229 (14.2)	221 (13.7)	0.014
ER- status	177 (3.7)	220 (2.9)	0.04	57 (3.5)	53 (3.3)	0.014
Comorbidities						
Overweight and obesity ^e	1979 (40.9)	1035 (13.8)	0.636	560 (34.8)	287 (17.8)	0.392
Bariatric surgery	94 (1.9)	101 (1.4)	0.046	28 (1.7)	29 (1.8)	0.005
CKD	242 (5.0)	146 (2.0)	0.167	51 (3.2)	67 (4.2)	0.053
COPD	151 (3.1)	185 (2.5)	0.039	51 (3.2)	47 (2.9)	0.014
Cerebrovascular diseases	171 (3.5)	188 (2.5)	0.059	43 (2.7)	45 (2.8)	0.008
Ischemic heart diseases	376 (7.8)	204 (2.7)	0.227	72 (4.5)	72 (4.5)	0.001
Diseases of arteries, arterioles and capillaries	271 (5.6)	181 (2.4)	<.001	58 (3.6)	57 (3.4)	0.003
Heart failure	233 (4.8)	105 (1.4)	0.197	44 (2.7)	36 (2.2)	0.032
Diseases of liver	438 (9.0)	220 (2.9)	0.259	83 (5.2)	88 (5.5)	0.014
Hypertensive diseases	2175 (44.9)	1701 (22.8)	0.481	507 (31.5)	514 (31.9)	0.009
Dyslipidemia	1949 (40.2)	1398 (18.7)	0.486	447 (27.8)	462 (28.7)	0.021
OSA	699 (14.4)	381 (5.1)	0.318	151 (9.4)	147 (9.1)	0.009

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Metformin	824 (17.0)	149 (2.0)	0.530	103 (6.4)	97 (6.0)	0.015
SGLT2i	229 (4.7)	11 (0.1)	0.300	13 (0.8)	11 (0.7)	0.014
Treatment						
Mastectomy Procedures	149 (3.1)	313 (4.2)	0.059	66 (4.1)	59 (3.7)	0.023
Mastectomy, partial ^f	97 (2.0)	182 (2.4)	0.029	42 (2.6)	35 (2.2)	0.028
Radiation therapy	136 (2.8)	143 (1.9)	0.059	42 (2.6)	40 (2.5)	0.008
Chemotherapy	1259 (26.0)	1195 (16.0)	0.248	359 (22.3)	357 (22.2)	0.003
Aromatase inhibitors	880 (18.2)	612 (8.2)	0.298	243 (15.1)	235 (14.6)	0.014
Tamoxifen	338 (7.0)	411 (5.5)	0.061	95 (5.9)	103 (6.4)	0.021
Taxanes	39 (0.8)	131 (1.8)	0.084	20 (1.2)	14 (0.9)	0.036
Platinum compounds	15 (0.3)	40 (0.5)	0.035	10 (0.6)	10 (0.6)	0.001
Anthracyclines and related substances	15 (0.3)	74 (1.0)	0.085	10 (0.6)	10 (0.6)	0.001
Cyclophosphamide	21 (0.4)	90 (1.2)	0.086	10 (0.6)	10 (0.6)	0.001

Abbreviations: GLP-1RA, glucagon-like peptide-1 receptor agonist; SGLT2i, sodium-glucose cotransporter 2 inhibitors, PSM, propensity score matching; SMD, standardized mean difference; ER+, estrogen receptor positive status; ER-, estrogen receptor negative status; CKD, chronic kidney disease; COPD, other chronic obstructive pulmonary disease; OSA, obstructive sleep apnea.

^aPropensity score matching for demographics, diagnosis, procedure, and medication.

^bSMD less than 0.10 indicates that the 2 comparison groups were well balanced.

^cOther race was defined as multiracial and any race not otherwise specified.

^dPersons with potential health hazards related to socioeconomic and psychosocial circumstances.

^eICD-10 code E66 for obesity is not universally applied across all health care organizations; thus, not all patients with body mass index 30 or higher (calculated as weight in kilograms divided by height in meters squared) will have a corresponding obesity diagnosis code E66.

^fMastectomy, partial (e.g., lumpectomy, tylectomy, quadrantectomy, segmentectomy).

eTable 3. Baseline Characteristics of GLP-1RA Use vs Insulin/Metformin Before and After Propensity Score Matching for All-Cause Mortality

Characteristic	Before PSM ^a			After PSM ^a		
	No. (%)		SMD ^b	No. (%)		SMD ^b
	GLP-1RA (n=2383)	Insulin/Metformin (n=51 397)		GLP-1RA (n=2323)	Insulin/Metformin (n=2323)	
Age at Index, mean (SD), y	63.5 (9.9)	66.9 (10.0)	0.347	63.7 (9.8)	63.6 (11.4)	0.005
Race						
American Indian or Alaska Native	10 (0.4)	202 (0.4)	0.004	10 (0.4)	10 (0.4)	0.001
Asian	103 (4.3)	2809 (5.5)	0.053	102 (4.4)	82 (3.5)	0.044
Black or African American	485 (20.4)	11676 (22.7)	0.058	473 (20.4)	459 (19.8)	0.015
Native Hawaiian or Other Pacific Islander	22 (0.9)	581 (1.1)	0.021	22 (0.9)	28 (1.2)	0.025
Other ^c	49 (2.1)	1732 (3.4)	0.081	49 (2.1)	45 (1.9)	0.012
Unknown race	78 (3.3)	2659 (5.2)	0.095	78 (3.4)	76 (3.3)	0.005
White	1637 (68.7)	31 738 (61.8)	0.146	1591 (68.5)	1625 (70.0)	0.032
Ethnicity						
Hispanic or Latino	129 (5.4)	4326 (8.4)	0.119	128 (5.5)	115 (5.0)	0.025
Not Hispanic or Latino	1727 (72.5)	38 226 (74.4)	0.043	1687 (72.6)	1714 (73.8)	0.026
Unknown Ethnicity	527 (22.1)	8845 (17.2)	0.124	508 (21.9)	494 (21.3)	0.015
Socioeconomic risks ^d	46 (1.9)	576 (1.1)	0.066	45 (1.9)	37 (1.6)	0.026
Lifestyle factors						
Nicotine dependence	94 (3.9)	2813 (5.5)	0.072	94 (4.0)	110 (4.7)	0.034
Alcohol related disorders	18 (0.8)	427 (0.8)	0.009	18 (0.8)	19 (0.8)	0.005
Breast cancer characteristics						
Carcinoma in situ of breast	209 (8.8)	4603 (9.0)	0.007	203 (8.7)	220 (9.5)	0.025
ER+ status	511 (21.4)	7082 (13.8)	0.202	495 (21.3)	467 (20.1)	0.03
ER- status	80 (3.4)	1726 (3.4)	0.001	79 (3.4)	71 (3.1)	0.019
Comorbidities						
Overweight and obesity ^e	1038 (43.6)	9721 (18.9)	0.552	997 (42.9)	607 (26.1)	0.359
Bariatric surgery	82 (3.4)	372 (0.7)	0.191	73 (3.1)	60 (2.6)	0.034
CKD	238 (10.0)	6129 (11.9)	0.062	236 (10.2)	244 (10.5)	0.011
COPD	107 (4.5)	3875 (7.5)	0.129	103 (4.4)	93 (4.0)	0.021
Cerebrovascular diseases	82 (3.4)	3767 (7.3)	0.173	81 (3.5)	64 (2.8)	0.042
Ischemic heart diseases	207 (8.7)	7168 (13.9)	0.167	202 (8.7)	201 (8.7)	0.002
Diseases of arteries, arterioles and capillaries	152 (6.4)	3914 (7.6)	0.048	150 (6.5)	167 (7.2)	0.029
Heart failure	138 (5.8)	5108 (9.9)	0.155	136 (5.9)	135 (5.8)	0.002
Diseases of liver	250 (10.5)	3335 (6.5)	0.144	241 (10.4)	233 (10.0)	0.011
Hypertensive diseases	1400 (58.7)	29 181 (56.8)	0.04	1358 (58.5)	1333 (57.4)	0.022
Dyslipidemia	1273 (53.4)	23 648 (46.0)	0.149	1227 (52.8)	1206 (51.9)	0.018
OSA	455 (19.1)	4022 (7.8)	0.335	415 (17.9)	391 (16.8)	0.027

Medications

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Metformin	26 (1.1)	12 461 (24.2)	0.743	26 (1.1)	28 (1.2)	0.008
SGLT2i	196 (8.2)	1167 (2.3)	0.269	183 (7.9)	174 (7.5)	0.015
Treatment						
Mastectomy Procedures	85 (3.6)	2359 (4.6)	0.052	85 (3.7)	80 (3.4)	0.012
Mastectomy, partial ^f	65 (2.7)	1580 (3.1)	0.021	65 (2.8)	64 (2.8)	0.003
Radiation therapy	67 (2.8)	1284 (2.5)	0.019	65 (2.8)	62 (2.7)	0.008
Chemotherapy	641 (26.9)	10 828 (21.1)	0.137	616 (26.5)	591 (25.4)	0.025
Aromatase inhibitors	465 (19.5)	6760 (13.2)	0.173	441 (19.0)	426 (18.3)	0.017
Tamoxifen	113 (4.7)	1474 (2.9)	0.098	105 (4.5)	104 (4.5)	0.002
Taxanes	15 (0.6)	984 (1.9)	0.115	15 (0.6)	16 (0.7)	0.005
Platinum compounds	10 (0.4)	346 (0.7)	0.034	10 (0.4)	10 (0.4)	0.001
Anthracyclines and related substances	10 (0.4)	569 (1.1)	0.079	10 (0.4)	10 (0.4)	0.001
Cyclophosphamide	10 (0.4)	707 (1.4)	0.101	10 (0.4)	10 (0.4)	0.001

Abbreviations: GLP-1RA, glucagon-like peptide-1 receptor agonist; SGLT2i, sodium-glucose cotransporter 2 inhibitors; PSM, propensity score matching; SMD, standardized mean difference; ER+, estrogen receptor positive status; ER-, estrogen receptor negative status; CKD, chronic kidney disease; COPD, other chronic obstructive pulmonary disease; OSA, obstructive sleep apnea.

^aPropensity score matching for demographics, diagnosis, procedure, and medication.

^bSMD less than 0.10 indicates that the 2 comparison groups were well balanced.

^cOther race was defined as multiracial and any race not otherwise specified.

^dPersons with potential health hazards related to socioeconomic and psychosocial circumstances.

^eICD-10 code E66 for obesity is not universally applied across all health care organizations; thus, not all patients with body mass index 30 or higher (calculated as weight in kilograms divided by height in meters squared) will have a corresponding obesity diagnosis code E66.

^fMastectomy, partial (e.g., lumpectomy, tylectomy, quadrantectomy, segmentectomy).

eTable 4. Baseline Characteristics of GLP-1RA Use vs SGLT2i Before and After Propensity Score Matching for All-Cause Mortality

Characteristic	Before PSM ^a			After PSM ^a		
	No. (%)		SMD ^b	No. (%)		SMD ^b
	GLP-1RA (n=11 689)	SGLT2i (n=6096)		GLP-1RA (n=4052)	SGLT2i (n=4052)	
Age at Index, mean (SD), y	63.4 (9.9)	69.9 (9.7)	0.662	68.6 (9.1)	68.5 (9.5)	0.005
Race						
American Indian or Alaska Native	54 (0.5)	18 (0.3)	0.027	19 (0.5)	12 (0.3)	0.028
Asian	426 (3.6)	492 (8.1)	0.189	288 (7.1)	282 (7.0)	0.006
Black or African American	2466 (21.1)	1472 (24.1)	0.073	945 (23.3)	938 (23.1)	0.004
Native Hawaiian or Other Pacific Islander	101 (0.9)	61 (1.0)	0.014	47 (1.2)	39 (1.0)	0.019
Other ^c	380 (3.3)	216 (3.5)	0.016	149 (3.7)	145 (3.6)	0.005
Unknown race	528 (4.5)	326 (5.3)	0.038	214 (5.3)	224 (5.5)	0.011
White	7734 (66.2)	3511 (57.6)	0.177	2390 (59.0)	2412 (59.5)	0.011
Ethnicity						
Hispanic or Latino	884 (7.6)	501 (8.2)	0.024	365 (9.0)	356 (8.8)	0.008
Not Hispanic or Latino	8721 (74.6)	4640 (76.1)	0.035	3056 (75.4)	3038 (75.0)	0.01
Unknown Ethnicity	2084 (17.8)	955 (15.7)	0.058	631 (15.6)	658 (16.2)	0.018
Socioeconomic risks ^d	190 (1.6)	189 (3.1)	0.097	75 (1.9)	79 (1.9)	0.007
Lifestyle factors						
Nicotine dependence	592 (5.1)	415 (6.8)	0.074	241 (5.9)	228 (5.6)	0.014
Alcohol related disorders	99 (0.8)	70 (1.1)	0.03	33 (0.8)	38 (0.9)	0.013
Breast cancer characteristics						
Carcinoma in situ of breast	1136 (9.7)	463 (7.6)	0.076	330 (8.1)	346 (8.5)	0.014
ER+ status	2224 (19.0)	1064 (17.5)	0.041	768 (19.0)	736 (18.2)	0.020
ER- status	413 (3.5)	209 (3.4)	0.006	135 (3.3)	144 (3.6)	0.012
Comorbidities						
Overweight and obesity ^e	4776 (40.9)	1650 (27.1)	0.294	1592 (39.3)	1034 (25.5)	0.297
Bariatric surgery	258 (2.2)	53 (0.9)	0.109	40 (1.0)	37 (0.9)	0.008
CKD	1314 (11.2)	1526 (25.0)	0.364	816 (20.1)	798 (19.7)	0.011
COPD	684 (5.9)	705 (11.6)	0.204	329 (8.1)	325 (8.0)	0.004
Cerebrovascular diseases	569 (4.9)	647 (10.6)	0.216	331 (8.2)	311 (7.7)	0.018
Ischemic heart diseases	1252 (10.7)	1647 (27.0)	0.426	765 (18.9)	763 (18.8)	0.001
Diseases of arteries, arterioles and capillaries	867 (7.4)	854 (14.0)	0.214	427 (10.7)	428 (10)	0.001
Heart failure	711 (6.1)	1881 (30.9)	0.674	653 (16.1)	643 (15.9)	0.007
Diseases of liver	1288 (11.0)	530 (8.7)	0.078	360 (8.9)	355 (8.8)	0.004
Hypertensive diseases	7392 (63.2)	4450 (73.0)	0.211	2897 (71.5)	2885 (71.2)	0.007
Dyslipidemia	6861 (58.7)	3974 (65.2)	0.134	2624 (64.8)	2597 (64.1)	0.014
OSA	2046 (17.5)	873 (14.3)	0.087	572 (14.1)	560 (13.8)	0.009

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Metformin	5030 (43.0)	2472 (40.6)	0.05	1687 (41.6)	1741 (43.0)	0.027
SGLT2i	39 (0.3)	948 (15.6)	0.586	39 (1.0)	52 (1.3)	0.030
Treatment						
Mastectomy Procedures	499 (4.3)	217 (3.6)	0.037	148 (3.7)	152 (3.8)	0.005
Mastectomy, partial ^f	342 (2.9)	144 (2.4)	0.035	92 (2.3)	99 (2.4)	0.011
Radiation therapy	360 (3.1)	154 (2.5)	0.034	113 (2.8)	110 (2.7)	0.005
Chemotherapy	3240 (27.7)	1845 (30.3)	0.056	1141 (28.2)	1172 (28.9)	0.017
Aromatase inhibitors	2354 (20.1)	1224 (20.1)	0.001	797 (19.7)	813 (20.1)	0.010
Tamoxifen	559 (4.8)	175 (2.9)	0.1	141 (3.5)	131 (3.2)	0.014
Taxanes	136 (1.2)	67 (1.1)	0.006	54 (1.3)	45 (1.1)	0.020
Platinum compounds	44 (0.4)	22 (0.4)	0.003	17 (0.4)	16 (0.4)	0.004
Anthracyclines and related substances	67 (0.6)	33 (0.5)	0.004	25 (0.6)	24 (0.6)	0.003
Cyclophosphamide	95 (0.8)	42 (0.7)	0.014	35 (0.9)	29 (0.7)	0.017

Abbreviations: GLP-1RA, glucagon-like peptide-1 receptor agonist; SGLT2i, sodium-glucose cotransporter 2 inhibitors, PSM, propensity score matching; SMD, standardized mean difference; ER+, estrogen receptor positive status; ER-, estrogen receptor negative status; CKD, chronic kidney disease; COPD, other chronic obstructive pulmonary disease; OSA, obstructive sleep apnea.

^aPropensity score matching for demographics, diagnosis, procedure, and medication.

^bSMD less than 0.10 indicates that the 2 comparison groups were well balanced.

^cOther race was defined as multiracial and any race not otherwise specified.

^dPersons with potential health hazards related to socioeconomic and psychosocial circumstances.

^eICD-10 code E66 for obesity is not universally applied across all health care organizations; thus, not all patients with body mass index 30 or higher (calculated as weight in kilograms divided by height in meters squared) will have a corresponding obesity diagnosis code E66.

^fMastectomy, partial (e.g., lumpectomy, tylectomy, quadrantectomy, segmentectomy).

eTable 5. Baseline Characteristics of GLP-1RA Use vs Nonuse Before and After Propensity Score Matching for Recurrence-Free Survival

Characteristic	Before PSM ^a			After PSM ^a		
	No. (%)		SMD ^b	No. (%)		SMD ^b
	GLP-1RA (n=4846)	Nonuse (n=7473)		GLP-1RA (n=1568)	Nonuse (n=1568)	
Age at Index, mean (SD), y	60.1 (10.7)	54.5 (11.4)	0.508	57.9 (10.7)	58.7 (11.2)	0.073
Race						
American Indian or Alaska Native	10 (0.2)	24 (0.3)	0.022	10 (0.6)	10 (0.6)	<0.001
Asian	94 (1.9)	62 (0.8)	0.095	24 (1.5)	23 (1.5)	0.005
Black or African American	731 (15.1)	1044 (14.0)	0.032	209 (13.3)	219 (14.0)	0.019
Native Hawaiian or Other Pacific Islander	15 (0.3)	23 (0.3)	<0.001	10 (0.6)	10 (0.6)	<0.001
Other ^c	143 (3.0)	225 (3.0)	0.003	58 (3.7)	59 (3.8)	0.003
Unknown race	269 (5.6)	411 (5.5)	0.002	96 (6.1)	88 (5.6)	0.022
White	3584 (74.0)	5684 (76.1)	0.048	1177 (75.1)	1173 (74.8)	0.006
Ethnicity						
Hispanic or Latino	306 (6.3)	468 (6.3)	0.002	108 (6.9)	99 (6.3)	0.023
Not Hispanic or Latino	3769 (77.8)	5576 (74.6)	0.074	1156 (73.7)	1176 (75.0)	0.029
Unknown Ethnicity	771 (15.9)	1429 (19.1)	0.085	304 (19.4)	293 (18.7)	0.018
Socioeconomic risks ^d	78 (1.6)	71 (1.0)	0.059	18 (1.1)	17 (1.1)	0.006
Lifestyle factors						
Nicotine dependence	133 (2.7)	336 (4.5)	0.094	52 (3.3)	39 (2.5)	0.049
Alcohol related disorders	37 (0.8)	129 (1.7)	0.087	22 (1.4)	16 (1.0)	0.035
Breast cancer characteristics						
Carcinoma in situ of breast	403 (8.3)	752 (10.1)	0.06	114 (7.3)	114 (7.3)	<0.001
ER+ status	935 (19.3)	826 (11.1)	0.231	231 (14.7)	206 (13.1)	0.046
ER- status	177 (3.7)	220 (2.9)	0.04	47 (3.0)	48 (3.1)	0.004
Comorbidities						
Overweight and obesity ^e	1981 (40.9)	1035 (13.8)	0.636	344 (21.9)	368 (23.5)	0.037
Bariatric surgery	94 (1.9)	101 (1.4)	0.046	19 (1.2)	26 (1.7)	0.038
CKD	242 (5.0)	146 (2.0)	0.167	53 (3.4)	50 (3.2)	0.011
COPD	151 (3.1)	185 (2.5)	0.039	49 (3.1)	46 (2.9)	0.011
Cerebrovascular diseases	171 (3.5)	188 (2.5)	0.059	40 (2.6)	39 (2.5)	0.004
Ischemic heart diseases	376 (7.8)	204 (2.7)	0.227	78 (5.0)	66 (4.2)	0.037
Diseases of arteries, arterioles and capillaries	271 (5.6)	181 (2.4)	0.162	58 (3.6)	57 (3.5)	0.003
Heart failure	233 (4.8)	105 (1.4)	0.197	44 (2.8)	36 (2.3)	0.032
Diseases of liver	1288 (11.0)	530 (8.7)	0.078	360 (8.9)	355 (8.8)	0.004
Hypertensive diseases	2177 (44.9)	1701 (22.8)	0.482	478 (30.5)	476 (30.4)	0.003
Dyslipidemia	1949 (40.2)	1398 (18.7)	0.486	447 (27.8)	462 (28.7)	0.021
OSA	700 (14.4)	380 (5.1)	0.319	135 (8.6)	141 (9.0)	0.014

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Metformin	824 (17.0)	149 (2.0)	0.53	101 (6.4)	97 (6.2)	0.01
SGLT2i	229 (4.7)	11 (0.1)	0.3	16 (1.0)	11 (0.7)	0.035

Treatment

Mastectomy Procedures	149 (3.1)	314 (4.2)	0.060	60 (3.8)	50 (3.2)	0.035
Mastectomy, partial ^f	97 (2.0)	182 (2.4)	0.114	38 (2.4)	35 (2.2)	0.013
Radiation therapy	136 (2.8)	143 (1.9)	0.059	37 (2.4)	31 (2.0)	0.026
Chemotherapy	1259 (26.0)	1196 (16.0)	0.247	342 (21.8)	337 (21.5)	0.008
Aromatase inhibitors	880 (18.2)	613 (8.2)	0.298	223 (14.2)	226 (14.4)	0.005
Tamoxifen	338 (7.0)	411 (5.5)	0.061	98 (6.3)	89 (5.7)	0.024
Taxanes	39 (0.8)	132 (1.8)	0.085	22 (1.4)	15 (1.0)	0.041
Platinum compounds	15 (0.3)	40 (0.5)	0.035	10 (0.6)	10 (0.6)	<0.001
Anthracyclines and related substances	15 (0.3)	74 (1.0)	0.085	10 (0.6)	10 (0.6)	<0.001
Cyclophosphamide	21 (0.4)	90 (1.2)	0.086	14 (0.9)	10 (0.6)	0.029

Abbreviations: GLP-1RA, glucagon-like peptide-1 receptor agonist; SGLT2i, sodium-glucose cotransporter 2 inhibitors; PSM, propensity score matching; SMD, standardized mean difference; ER+, estrogen receptor positive status; ER-, estrogen receptor negative status; CKD, chronic kidney disease; COPD, other chronic obstructive pulmonary disease; OSA, obstructive sleep apnea.

^aPropensity score matching for demographics, diagnosis, procedure, and medication.

^bSMD less than 0.10 indicates that the 2 comparison groups were well balanced.

^cOther race was defined as multiracial and any race not otherwise specified.

^dPersons with potential health hazards related to socioeconomic and psychosocial circumstances.

^eICD-10 code E66 for obesity is not universally applied across all health care organizations; thus, not all patients with body mass index 30 or higher (calculated as weight in kilograms divided by height in meters squared) will have a corresponding obesity diagnosis code E66.

^fMastectomy, partial (e.g., lumpectomy, tylectomy, quadrantectomy, segmentectomy).

eTable 6. Baseline Characteristics of GLP-1RA Use vs Insulin/Metformin Before and After Propensity Score Matching for Recurrence-Free Survival

Characteristic	Before PSM ^a			After PSM ^a		
	No. (%)		SMD ^b	No. (%)		SMD ^b
	GLP-1RA (n=2383)	Insulin/Metformin (n=51 397)		GLP-1RA (n=2312)	Insulin/ Metformin (n=2312)	
Age at Index, mean (SD), y	63.5 (9.9)	66.9 (10.0)	0.347	63.7 (9.7)	63.2 (11.5)	0.05
Race						
American Indian or Alaska Native	10 (0.4)	202 (0.4)	0.004	10 (0.4)	10 (0.4)	<0.001
Asian	103 (4.3)	2809 (5.5)	0.053	100 (4.3)	94 (4.1)	0.013
Black or African American	485 (20.4)	11676 (22.7)	0.058	473 (20.5)	471 (20.4)	0.002
Native Hawaiian or Other Pacific Islander	22 (0.9)	581 (1.1)	0.021	22 (1.0)	34 (1.5)	0.047
Other ^c	49 (2.1)	1732 (3.4)	0.081	49 (2.1)	50 (2.2)	0.003
Unknown race	78 (3.3)	2659 (5.2)	0.095	78 (3.4)	71 (3.1)	0.017
White	1637 (68.7)	31738 (61.8)	0.146	1581 (68.4)	1586 (68.6)	0.005
Ethnicity						
Hispanic or Latino	129 (5.4)	4326 (8.4)	0.119	127 (5.5)	152 (6.6)	0.045
Not Hispanic or Latino	1727 (72.5)	38226 (74.4)	0.043	1678 (72.6)	1675 (72.4)	0.003
Unknown Ethnicity	527 (22.1)	8845 (17.2)	0.124	507 (21.9)	485 (21.0)	0.023
Socioeconomic risks ^d	46 (1.9)	576 (1.1)	0.066	44 (1.9)	37 (1.6)	0.023
Lifestyle factors						
Nicotine dependence	94 (3.9)	2813 (5.5)	0.072	94 (4.1)	90 (3.9)	0.009
Alcohol related disorders	18 (0.8)	427 (0.8)	0.009	18 (0.8)	19 (0.8)	0.005
Breast cancer characteristics						
Carcinoma in situ of breast	209 (8.8)	4603 (9.0)	0.007	203 (8.8)	200 (8.7)	0.005
ER+ status	511 (21.4)	7082 (13.8)	0.202	493 (21.3)	494 (21.4)	0.001
ER- status	80 (3.4)	1726 (3.4)	<0.001	79 (3.4)	71 (3.1)	0.020
Comorbidities						
Overweight and obesity ^e	1038 (43.6)	9721 (18.9)	0.552	974 (42.1)	934 (40.4)	0.035
Bariatric surgery	82 (3.4)	372 (0.7)	0.191	70 (3.0)	63 (2.7)	0.018
CKD	238 (10.0)	6129 (11.9)	0.062	235 (10.2)	211 (9.1)	0.035
COPD	107 (4.5)	3875 (7.5)	<0.001	104 (4.5)	99 (4.3)	0.011
Cerebrovascular diseases	82 (3.4)	3767 (7.3)	0.173	81 (3.5)	82 (3.5)	0.002
Ischemic heart diseases	207 (8.7)	7168 (13.9)	0.167	205 (8.9)	213 (9.2)	0.012
Diseases of arteries, arterioles and capillaries	152 (6.4)	3914 (7.6)	0.048	150 (6.5)	167 (7.2)	0.029
Heart failure	250 (10.5)	3335 (6.5)	0.144	241 (10.4)	233 (10.0)	0.011
Diseases of liver	1400 (58.7)	29181 (56.8)	0.040	1352 (58.5)	1310 (56.7)	0.037
Hypertensive diseases	1273 (53.4)	23 648 (46.0)	0.149	1227 (53.1)	1195 (51.7)	0.028
Dyslipidemia	455 (19.1)	4022 (7.8)	0.335	412 (17.8)	372 (16.1)	0.046
OSA	152 (6.4)	3914 (7.6)	0.048	150 (6.5)	167 (7.2)	0.029

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Metformin	26 (1.1)	12 461 (24.2)	0.743	26 (1.1)	33 (1.4)	0.027
SGLT2i	196 (8.2)	1167 (2.3)	0.269	182 (7.9)	178 (7.7)	0.006
Treatment						
Mastectomy Procedures	85 (3.6)	2359 (4.6)	0.052	85 (3.7)	85 (3.7)	<0.001
Mastectomy, partial ^f	65 (2.7)	1580 (3.1)	0.021	65 (2.8)	63 (2.7)	0.005
Radiation therapy	67 (2.8)	1284 (2.5)	0.019	65 (2.8)	60 (2.6)	0.013
Chemotherapy	641 (26.9)	10828 (21.1)	0.137	616 (26.6)	647 (28.0)	0.030
Aromatase inhibitors	465 (19.5)	6760 (13.2)	0.173	442 (19.1)	461 (19.9)	0.021
Tamoxifen	113 (4.7)	1474 (2.9)	0.098	104 (4.5)	118 (5.1)	0.028
Taxanes	15 (0.6)	984 (1.9)	0.115	15 (0.6)	14 (0.6)	0.005
Platinum compounds	10 (0.4)	346 (0.7)	0.034	10 (0.4)	<10 (1.0)	<0.001
Anthracyclines and related substances	10 (0.4)	569 (1.1)	0.079	10 (0.4)	<10 (1.0)	<0.001
Cyclophosphamide	10 (0.4)	707 (1.4)	0.101	10 (0.4)	<10 (1.0)	<0.001

Abbreviations: GLP-1RA, glucagon-like peptide-1 receptor agonist; SGLT2i, sodium-glucose cotransporter 2 inhibitors, PSM, propensity score matching; SMD, standardized mean difference; ER+, estrogen receptor positive status; ER-, estrogen receptor negative status; CKD, chronic kidney disease; COPD, other chronic obstructive pulmonary disease; OSA, obstructive sleep apnea.

^aPropensity score matching for demographics, diagnosis, procedure, and medication.

^bSMD less than 0.10 indicates that the 2 comparison groups were well balanced.

^cOther race was defined as multiracial and any race not otherwise specified.

^dPersons with potential health hazards related to socioeconomic and psychosocial circumstances.

^eICD-10 code E66 for obesity is not universally applied across all health care organizations; thus, not all patients with body mass index 30 or higher (calculated as weight in kilograms divided by height in meters squared) will have a corresponding obesity diagnosis code E66.

^fMastectomy, partial (e.g., lumpectomy, tylectomy, quadrantectomy, segmentectomy).

eTable 7. Baseline Characteristics of GLP-1RA Use vs SGLT2i Before and After Propensity Score Matching for Recurrence-Free Survival

Characteristic	Before PSM ^a			After PSM ^a		
	No. (%)		SMD ^b	No. (%)		SMD ^b
	GLP-1RA (n=11 689)	SGLT2i (n=6096)		GLP-1RA (n=4000)	SGLT2i (n=4000)	
Age at Index, mean (SD), y	63.4 (9.9)	69.9 (9.7)	0.662	68.4 (9.2)	68.4 (9.5)	<0.001
Race						
American Indian or Alaska Native	54 (0.5)	18 (0.3)	0.027	13 (0.3)	12 (0.3)	0.004
Asian	426 (3.6)	492 (8.1)	0.189	285 (7.1)	280 (7.0)	0.005
Black or African American	2466 (21.1)	1472 (24.1)	0.073	924 (23.1)	930 (23.3)	0.004
Native Hawaiian or Other Pacific Islander	101 (0.9)	61 (1.0)	0.014	39 (1.0)	40 (1.0)	0.003
Other ^c	380 (3.3)	216 (3.5)	0.016	152 (3.8)	146 (3.7)	0.008
Unknown race	528 (4.5)	326 (5.3)	0.038	226 (5.7)	214 (5.4)	0.013
White	7734 (66.2)	3511 (57.6)	0.177	2361 (59.0)	2378 (59.5)	0.009
Ethnicity						
Hispanic or Latino	884 (7.6)	501 (8.2)	0.024	359 (9.0)	351 (8.8)	0.007
Not Hispanic or Latino	8721 (74.6)	4640 (76.1)	0.035	3029 (75.7)	3004 (75.1)	0.015
Unknown Ethnicity	2084 (17.8)	955 (15.7)	0.058	612 (15.3)	645 (16.1)	0.023
Socioeconomic risks ^d	190 (1.6)	189 (3.1)	0.097	71 (1.8)	78 (2.0)	0.013
Lifestyle factors						
Nicotine dependence	592 (5.1)	415 (6.8)	0.074	226 (5.7)	227 (5.7)	0.001
Alcohol related disorders	99 (0.8)	70 (1.1)	0.030	35 (0.9)	35 (0.9)	<0.001
Breast cancer characteristics						
Carcinoma in situ of breast	1136 (9.7)	463 (7.6)	0.076	325 (8.1)	341 (8.5)	0.014
ER+ status	2224 (19.0)	1064 (17.5)	0.041	738 (18.5)	726 (18.2)	0.008
ER- status	413 (3.5)	209 (3.4)	0.006	127 (3.2)	143 (3.6)	0.022
Comorbidities						
Overweight and obesity ^e	4 776 (40.9)	1650 (27.1)	0.294	1107 (27.7)	1131 (28.3)	0.013
Bariatric surgery	258 (2.2)	53 (0.9)	0.109	38 (1.0)	39 (1.0)	0.003
CKD	1314 (11.2)	1526 (25.0)	0.364	775 (19.4)	788 (19.7)	0.008
COPD	684 (5.9)	705 (11.6)	0.204	320 (8.0)	324 (8.1)	0.004
Cerebrovascular diseases	569 (4.9)	647 (10.6)	0.216	317 (7.9)	310 (7.8)	0.007
Ischemic heart diseases	1252 (10.7)	1647 (27.0)	0.426	743 (18.6)	739 (18.5)	0.003
Diseases of arteries, arterioles and capillaries	867 (7.4)	854 (14.0)	0.214	427 (10.7)	428 (10)	0.001
Heart failure	711 (6.1)	1881 (30.9)	0.674	635 (15.9)	646 (16.2)	0.007
Diseases of liver	1288 (11.0)	530 (8.7)	0.078	371 (9.3)	349 (8.7)	0.019
Hypertensive diseases	7392 (63.2)	4450 (73.0)	0.211	2826 (70.7)	2840 (71.0)	0.008
Dyslipidemia	6861 (58.7)	3974 (65.2)	0.134	2549 (63.7)	2561 (64.0)	0.006
OSA	2046 (17.5)	873 (14.3)	0.087	560 (14.0)	562 (14.1)	0.001

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Metformin	5030 (43.0)	2472 (40.6)	0.05	1710 (42.8)	1711 (42.8)	0.001
SGLT2i	39 (0.3)	948 (15.6)	0.586	39 (1.0)	45 (1.1)	0.015
Treatment						
Mastectomy Procedures	499 (4.3)	217 (3.6)	0.037	157 (3.9)	152 (3.8)	0.006
Mastectomy, partial ^f	342 (2.9)	144 (2.4)	0.035	99 (2.5)	101 (2.5)	0.003
Radiation therapy	360 (3.1)	154 (2.5)	0.034	121 (3.0)	108 (2.7)	0.019
Chemotherapy	3240 (27.7)	1845 (30.3)	0.056	1154 (28.8)	1163 (29.1)	0.005
Aromatase inhibitors	2354 (20.1)	1224 (20.1)	0.001	809 (20.2)	809 (20.2)	<0.001
Tamoxifen	559 (4.8)	175 (2.9)	0.1	143 (3.6)	132 (3.3)	0.015
Taxanes	136 (1.2)	67 (1.1)	0.006	42 (1.1)	46 (1.2)	0.010
Platinum compounds	44 (0.4)	22 (0.4)	0.003	18 (0.4)	16 (0.4)	0.008
Anthracyclines and related substances	67 (0.6)	33 (0.5)	0.004	23 (0.6)	22 (0.5)	0.003
Cyclophosphamide	95 (0.8)	42 (0.7)	0.014	26 (0.7)	28 (0.7)	0.006

Abbreviations: GLP-1RA, glucagon-like peptide-1 receptor agonist; SGLT2i, sodium-glucose cotransporter 2 inhibitors; PSM, propensity score matching; SMD, standardized mean difference; ER+, estrogen receptor positive status; ER-, estrogen receptor negative status; CKD, chronic kidney disease; COPD, other chronic obstructive pulmonary disease; OSA, obstructive sleep apnea.

^aPropensity score matching for demographics, diagnosis, procedure, and medication.

^bSMD less than 0.10 indicates that the 2 comparison groups were well balanced.

^cOther race was defined as multiracial and any race not otherwise specified.

^dPersons with potential health hazards related to socioeconomic and psychosocial circumstances.

^eICD-10 code E66 for obesity is not universally applied across all health care organizations; thus, not all patients with body mass index 30 or higher (calculated as weight in kilograms divided by height in meters squared) will have a corresponding obesity diagnosis code E66.

^fMastectomy, partial (e.g., lumpectomy, tylectomy, quadrantectomy, segmentectomy).

eTable 8. Mean, Median, Standard Deviation, and Interquartile Range of Follow-Up Time (in Days) of Three Comparisons Over 10-Year Follow Ups for All-Cause Mortality and Recurrence-Free Survival Outcomes Among Patients with Breast Cancer

	Before PSM ^a				After PSM ^a			
	Mean Follow-up (Days)	SD	Median Follow-up (Days)	IQR ^b	Mean Follow-up (Days)	SD	Median Follow-up (Days)	IQR ^b
All-cause mortality								
GLP-1RA use	657	544	553	579	962	694	819	659
Nonuse	2137	1172	2120	2224	997	708	931	832
GLP-1RA use	721	565	613	646	730	568	625	646
Insulin or metformin	1620	1167	1396	1916	663	590	525	737
GLP-1RA use	1049	826	839	943	988	776	806	902
SGLT2i use	898	739	729	855	963	772	776	911
Recurrence-free survival								
GLP-1RA use	656	544	552	580	951	703	806	667
Nonuse	2137	1172	2120	2227	1000	704	934	865
GLP-1RA use	721	565	613	646	731	568	625	644
insulin or metformin	1620	1167	1396	1916	683	630	532	754
GLP-1RA use	1049	826	839	943	991	781	804	898
SGLT2i	898	739	729	855	963	772	774	906

Abbreviations: PSM, propensity score matching; SD, standard deviation; IQR, interquartile range; GLP-1RA, glucagon-like peptide-1 receptor agonists; SGLT2i, sodium-glucose cotransporter 2 inhibitors; BC, breast cancer; T2D, type 2 diabetes.

^aPropensity score matching for demographics, diagnosis, procedure, and medication.

^bTriNetx reports the IQR as a single value.

eTable 9. Cox Proportional Hazards Regression Models for the Association of GLP-1RA Use vs Insulin or Metformin vs SGLT2i Over 10-Year Follow-Up for All-Cause Mortality

	GLP-1RA vs Nonuse		GLP-1RA vs Insulin or Metformin		GLP-1RA vs SGLT2i	
	HR (95% CI)	P value	HR (95% CI)	P value	HR (95% CI)	P value
GLP-1RA exposure	0.42 (0.28–0.62)	<.001	0.12 (0.08–0.18)	<.001	0.77 (0.68–0.88)	<.001
Age at Index, mean (SD), y	1.04 (1.03–1.05)	<.001	1.04 (1.04–1.04)	<.001	1.04 (1.04–1.05)	<.001
Race						
American Indian or Alaska Native	1.48 (0.34–6.43)	0.603	0.86 (0.61–1.20)	0.372	1.07 (0.56–2.06)	0.84
Asian	0.62 (0.14–2.70)	0.526	0.61 (0.52–0.71)	<.001	0.48 (0.31–0.73)	<.001
Black or African American	1.08 (0.63–1.83)	0.789	0.83 (0.74–0.93)	0.001	0.58 (0.43–0.79)	<.001
Native Hawaiian or Other Pacific Islander	0.79 (0.10–5.93)	0.816	0.95 (0.78–1.17)	0.642	0.61 (0.33–1.15)	0.12
Other race ^a	0.91 (0.39–2.12)	0.829	0.99 (0.83–1.17)	0.871	0.91 (0.58–1.42)	0.68
White	0.95 (0.59–1.52)	0.82	0.98 (0.88–1.10)	0.755	0.77 (0.58–1.01)	0.06
Ethnicity						
Hispanic or Latino	0.60 (0.33–1.08)	0.088	0.69 (0.62–0.77)	<.001	0.57 (0.41–0.80)	0.001
Not Hispanic or Latino	0.80 (0.62–1.02)	0.070	0.96 (0.91–1.01)	0.121	1.03 (0.88–1.21)	0.71
Socioeconomic risks ^b	0.62 (0.29–1.33)	0.218	0.97 (0.84–1.10)	0.602	0.89 (0.70–1.14)	0.37
Comorbidities						
Overweight and obesity ^c	0.86 (0.67–1.11)	0.245	1.01 (0.96–1.06)	0.749	0.96 (0.84–1.09)	0.51
Bariatric surgery status	0.81 (0.40–1.66)	0.573	0.94 (0.77–1.16)	0.563	1.00 (0.67–1.49)	0.98
CKD	1.86 (1.23–2.81)	0.003	1.73 (1.64–1.82)	<.001	1.53 (1.34–1.75)	<.001
COPD	1.64 (1.13–2.39)	0.01	1.40 (1.32–1.49)	<.001	1.34 (1.16–1.56)	<.001
Cerebrovascular diseases	1.44 (1.02–2.05)	0.04	1.29 (1.22–1.37)	<.001	1.30 (1.13–1.49)	<.001
Ischemic heart diseases	0.85 (0.58–1.25)	0.417	1.21 (1.15–1.27)	<.001	1.34 (1.16–1.54)	<.001
Heart failure	1.97 (1.29–3.00)	0.002	1.82 (1.72–1.93)	<.001	2.40 (2.07–2.78)	<.001
Diseases of arteries, arterioles, capillaries	1.05 (0.73–1.51)	0.787	1.12 (1.06–1.18)	<.001	1.13 (0.98–1.29)	0.08
Diseases of liver	1.47 (1.07–2.02)	0.017	1.25 (1.18–1.33)	<.001	1.05 (0.91–1.20)	0.50
Hypertensive diseases	0.91 (0.72–1.16)	0.446	0.99 (0.93–1.04)	0.615	0.88 (0.72–1.08)	0.23
Disorders of lipoprotein metabolism	0.82 (0.64–1.04)	0.106	0.69 (0.66–0.72)	<.001	0.71 (0.59–0.84)	<.001
OSA	1.23 (0.90–1.68)	0.197	0.92 (0.87–0.98)	0.012	0.94 (0.82–1.07)	0.34
Lifestyle factors						
Alcohol related disorders	1.40 (0.88–2.25)	0.156	1.17 (1.02–1.35)	0.029	1.42 (1.04–1.93)	0.03
Nicotine dependence	1.35 (0.97–1.88)	0.076	1.28 (1.20–1.37)	<.001	1.39 (1.18–1.64)	<.001
Diagnosis						
Carcinoma in situ of breast	0.56 (0.40–0.80)	0.001	0.74 (0.69–0.79)	<.001	0.72 (0.61–0.85)	<.001
ER+ status	1.19 (0.87–1.63)	0.274	0.89 (0.83–0.95)	<.001	0.89 (0.77–1.05)	0.16
ER- status	0.81 (0.47–1.38)	0.439	1.05 (0.94–1.17)	0.354	1.14 (0.89–1.46)	0.31
Stage 0	0.37 (0.09–1.51)	0.165	0.96 (0.78–1.19)	0.733	1.23 (0.72–2.08)	0.45
Stage 1	0.47 (0.21–1.05)	0.067	0.83 (0.71–0.97)	0.023	0.89 (0.57–1.38)	0.60
Stage 2	0.61 (0.26–1.42)	0.248	1.01 (0.85–1.21)	0.872	0.72 (0.40–1.27)	0.25
Stage 3	2.26 (0.92–5.56)	0.077	1.61 (1.28–2.03)	<.001	2.08 (1.12–3.87)	0.02
HER2-	1.98 (1.03–3.79)	0.039	0.96 (0.82–1.13)	0.625	0.96 (0.61–1.50)	0.86
HER2+	1.27 (0.53–3.04)	0.584	0.74 (0.59–0.92)	0.006	0.62 (0.34–1.14)	0.13
Medications						
Insulins and analogues	1.79 (1.25–2.56)	0.001	1.27 (1.21–1.32)	<.001	1.63 (1.43–1.85)	<.001
Metformin	1.79 (1.20–2.69)	0.005	0.70 (0.67–0.73)	<.001	0.86 (0.76–0.97)	0.01
SGLT2i	0.99 (0.43–2.29)	0.983	0.93 (0.81–1.06)	0.284	1.36 (1.13–1.63)	0.001
Treatment						

Mastectomy procedures	0.44 (0.22–0.86)	0.017	1.09 (0.97–1.23)	0.138	0.96 (0.75–1.23)	0.74
Mastectomy, partial ^d	1.93 (0.88–4.21)	0.101	0.73 (0.64–0.84)	<.001	0.78 (0.59–1.02)	0.07
Chemotherapy	1.59 (1.18–2.15)	0.003	1.24 (1.15–1.32)	<.001	1.15 (0.97–1.37)	0.11
Tamoxifen	0.86 (0.59–1.25)	0.427	0.86 (0.78–0.95)	0.003	1.12 (0.91–1.38)	0.27
Aromatase inhibitors	0.78 (0.54–1.12)	0.179	0.98 (0.90–1.06)	0.558	1.00 (0.83–1.20)	0.99
Platinum compounds	3.04 (1.09–8.49)	0.033	1.78 (1.43–2.22)	<.001	1.03 (0.59–1.80)	0.92
Anthracyclines	2.05 (0.87–4.81)	0.100	1.41 (1.16–1.71)	0.001	1.30 (0.85–2.00)	0.22
Cyclophosphamide	1.48 (0.53–4.15)	0.451	0.72 (0.58–0.89)	0.002	0.67 (0.40–1.11)	0.12
Taxanes	0.51 (0.23–1.14)	0.102	1.12 (0.94–1.33)	0.215	1.33 (0.86–2.06)	0.20

Abbreviations: GLP-1RA, glucagon-like peptide-1 receptor agonist; SGLT2i, sodium-glucose cotransporter 2 inhibitors, ER+, estrogen receptor positive status; ER-, estrogen receptor negative status; CKD, chronic kidney disease; COPD, other chronic obstructive pulmonary disease; OSA, obstructive sleep apnea; HR, Hazard Ratio; HER2+, Human epidermal growth factor receptor 2-positive; and HER2-, Human epidermal growth factor receptor 2-negative

^aOther race was defined as multiracial and any race not otherwise specified.

^bPersons with potential health hazards related to socioeconomic and psychosocial circumstances.

group, including family, problems related to certain psychosocial circumstances, and problems related to other psychosocial circumstances.

^cICD-10 code E66 for obesity is not universally applied across all health care organizations; thus, not all patients with body mass index 30 or higher (calculated as weight in kilograms divided by height in meters squared) will have a corresponding obesity diagnosis code E66.

^dMastectomy, partial (e.g., lumpectomy, tylectomy, quadrantectomy, segmentectomy).

eTable 10. Cox Proportional Hazards Regression Models for the Association of GLP-1RA Use vs Insulin or Metformin vs SGLT2i Over 10-Year Follow-Up for Recurrence-Free Survival

	GLP-1RA vs Nonuse HR (95% CI)	<i>P</i> value	GLP-1 RA vs Insulin/Metformin HR (95% CI)	<i>P</i> value	GLP-1 RA vs SGLT2i HR (95% CI)	<i>P</i> value
GLP-1RA exposure	0.56 (0.46–0.67)	<.001	0.44 (0.36–0.54)	<.001	0.84 (0.73–0.97)	0.02
Age at Index, mean (SD), y	0.98 (0.98–0.99)	<.001	0.99 (0.99–1.00)	<.001	0.99 (0.99–1.00)	0.05
Race						
American Indian or Alaska Native	0.23 (0.03–1.69)	0.15	1.26 (0.87–1.84)	0.224	0.92 (0.33–2.57)	0.87
Asian	0.62 (0.26–1.47)	0.28	1.23 (1.03–1.46)	0.022	1.54 (1.02–2.33)	0.04
Black or African American	1.45 (1.03–2.02)	0.03	1.23 (1.07–1.42)	0.004	1.40 (0.99–1.99)	0.06
Native Hawaiian or Other Pacific Islander	3.00 (1.34–6.70)	0.01	2.11 (1.67–2.67)	<.001	1.82 (0.98–3.40)	0.06
Other race ^a	1.81 (1.16–2.81)	0.01	1.28 (1.06–1.54)	0.012	1.49 (0.95–2.32)	0.08
White	1.16 (0.85–1.58)	0.35	1.24 (1.08–1.41)	0.002	1.37 (0.98–1.90)	0.07
Ethnicity						
Hispanic or Latino	0.86 (0.63–1.17)	0.33	0.83 (0.74–0.93)	<.001	1.00 (0.77–1.29)	0.98
Not Hispanic or Latino	1.03 (0.88–1.22)	0.68	0.76 (0.71–0.81)	<.001	0.88 (0.76–1.03)	0.12
Socioeconomic risks ^b	1.24 (0.89–1.74)	0.21	1.34 (1.16–1.54)	<.001	0.96 (0.75–1.22)	0.73
Comorbidities						
Overweight and obesity ^c	1.05 (0.90–1.22)	0.53	1.07 (1.01–1.14)	0.029	0.97 (0.85–1.11)	0.69
Bariatric surgery status	0.93 (0.60–1.45)	0.76	0.83 (0.64–1.08)	0.172	1.26 (0.91–1.73)	0.16
CKD	0.74 (0.48–1.13)	0.17	1.02 (0.94–1.11)	0.598	1.03 (0.88–1.21)	0.72
COPD	1.26 (0.90–1.76)	0.18	0.99 (0.90–1.09)	0.888	1.16 (0.95–1.40)	0.14
Cerebrovascular diseases	0.93 (0.68–1.27)	0.63	1.01 (0.92–1.10)	0.864	0.83 (0.69–1.01)	0.06
Ischemic heart diseases	1.04 (0.79–1.37)	0.80	0.99 (0.92–1.06)	0.734	1.09 (0.93–1.27)	0.31
Heart failure	1.18 (0.83–1.68)	0.35	0.97 (0.88–1.06)	0.485	1.01 (0.84–1.20)	0.95
Diseases of liver	1.03 (0.83–1.27)	0.82	1.20 (1.11–1.29)	<.001	1.18 (1.03–1.36)	0.02
Hypertensive diseases	1.03 (0.89–1.20)	0.66	0.96 (0.90–1.02)	0.175	0.99 (0.84–1.16)	0.88
Disorders of lipoprotein metabolism	0.80 (0.69–0.94)	0.01	0.80 (0.75–0.85)	<.001	0.79 (0.68–0.91)	0.002
OSA	0.71 (0.56–0.90)	<.001	0.85 (0.77–0.93)	<.001	0.90 (0.78–1.05)	0.18
Lifestyle factors						

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Alcohol related disorders	0.69 (0.47–1.03)	0.07	1.05 (0.88–1.26)	0.589	0.85 (0.57–1.27)	0.44
Nicotine dependence	0.93 (0.74–1.17)	0.55	1.24 (1.14–1.35)	<.001	1.09 (0.91–1.31)	0.34
Diagnosis						
Carcinoma in situ of breast	0.75 (0.63–0.90)	<.001	0.72 (0.66–0.78)	<.001	0.81 (0.69–0.94)	0.01
ER+ status	2.04 (1.73–2.39)	<.001	2.06 (1.93–2.20)	<.001	2.12 (1.85–2.43)	<.001
ER- status	1.09 (0.85–1.40)	0.48	1.39 (1.25–1.54)	<.001	1.22 (0.99–1.49)	0.06
Stage 0	0.34 (0.16–0.72)	0.01	0.54 (0.40–0.71)	<.001	0.68 (0.39–1.19)	0.18
Stage 1	0.45 (0.30–0.68)	<.001	0.57 (0.48–0.68)	<.001	0.55 (0.38–0.78)	0.001
Stage 2	1.21 (0.86–1.71)	0.28	1.06 (0.92–1.23)	0.43	1.22 (0.86–1.72)	0.26
Stage 3	1.99 (1.27–3.12)	<.001	2.40 (2.02–2.84)	<.001	2.60 (1.76–3.82)	<.001
HER2-	1.81 (1.31–2.50)	<.001	1.30 (1.13–1.51)	<.001	1.34 (0.97–1.84)	0.07
HER2+	1.11 (0.76–1.61)	0.59	0.94 (0.80–1.10)	0.442	1.09 (0.78–1.52)	0.63
Medications						
Insulins and analogues	0.93 (0.71–1.23)	0.61	1.26 (1.19–1.33)	<.001	1.25 (1.11–1.42)	<.001
Metformin	0.97 (0.75–1.26)	0.81	1.09 (1.03–1.15)	0.003	0.93 (0.82–1.05)	0.26
SGLT2i	1.48 (0.86–2.56)	0.16	1.15 (1.00–1.33)	0.057	1.84 (1.54–2.21)	<.001
Treatment						
Mastectomy procedures	1.00 (0.79–1.27)	0.99	0.95 (0.84–1.06)	0.35	0.92 (0.75–1.12)	0.40
Mastectomy, partial ^d	0.50 (0.36–0.71)	<.001	0.56 (0.48–0.65)	<.001	0.69 (0.55–0.86)	0.001
Chemotherapy	1.51 (1.26–1.81)	<.001	1.66 (1.54–1.80)	<.001	1.44 (1.21–1.72)	<.001
Tamoxifen	0.72 (0.59–0.88)	<.001	0.63 (0.56–0.72)	<.001	0.83 (0.69–1.00)	0.06
Aromatase inhibitors	0.95 (0.78–1.17)	0.66	0.71 (0.65–0.78)	<.001	0.88 (0.74–1.05)	0.16
Platinum compounds	1.18 (0.76–1.81)	0.47	1.44 (1.20–1.74)	<.001	1.94 (1.39–2.69)	<.001
Anthracyclines	1.97 (1.33–2.93)	<.001	1.55 (1.30–1.83)	<.001	1.54 (1.13–2.09)	0.01
Cyclophosphamide	0.72 (0.44–1.20)	0.21	0.83 (0.69–1.02)	0.072	1.32 (0.90–1.94)	0.16
Taxanes	1.34 (0.90–1.99)	0.15	1.42 (1.21–1.66)	<.001	1.06 (0.76–1.47)	0.73

Abbreviations: GLP-1RA, glucagon-like peptide-1 receptor agonist; SGLT2i, sodium-glucose cotransporter 2 inhibitors; ER+, estrogen receptor positive status; ER-, estrogen receptor negative status; CKD, chronic kidney disease; COPD, other chronic obstructive pulmonary disease; OSA, obstructive sleep apnea; HR, Hazard Ratio; HER2+, Human epidermal growth factor receptor 2-positive; and HER2-, Human epidermal growth factor receptor 2-negative

^aOther race was defined as multiracial and any race not otherwise specified.

^bPersons with potential health hazards related to socioeconomic and psychosocial circumstances.

group, including family, problems related to certain psychosocial circumstances, and problems related to other psychosocial circumstances.

^cICD-10 code E66 for obesity is not universally applied across all health care organizations; thus, not all patients with body mass index 30 or higher (calculated as weight in kilograms divided by height in meters squared) will have a corresponding obesity diagnosis code E66.

^dMastectomy, partial (e.g., lumpectomy, tylectomy, quadrantectomy, segmentectomy).